

CCDP Traceability Matrix v0.1

Child-c Development Protocol — corpus dependency, control-surface,
and duplication audit

Draft Traceability / Corpus-Control Companion

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Bruxelles, 2026

Version: v0.1 | Date: 2026-05-13

Contents

CCDP Traceability Matrix v0.1	3
Child-c Development Protocol — corpus dependency, control-surface, and duplication audit	3
0. Purpose	3
1. Traceability discipline	3
1.1 Reading rule	3
1.2 Dependency relation types	3
1.3 Coverage states	4
2. Parent corpus registry	4
3. CCDP section-to-corpus matrix	5
4. Foundational invariant trace	7
5. Mechanism-to-evidence matrix	7
6. Duplication and stop-rule audit	8
7. Required witness families for CCDP	8
8. Open gaps and recommended child-specific modules	9
9. Corpus-risk register	9
10. Acceptance checklist for CCDP v0.1-R	10
11. Recommended next work order	10
12. Final traceability verdict	11

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Child-c Development Protocol — corpus dependency, control-surface, and duplication audit

Status: Draft traceability / corpus-control companion

Version: v0.1

Date: 2026-05-13

Language: English

Primary subject: CCDP_v0_1_R_Corpus_Aligned.md

Document class: corpus-control / traceability matrix

Assertion class: C-A10 for this matrix; mapped CCDP claims remain C-A4 unless a stronger parent layer is explicitly inherited

Precedence: this matrix does not override parent protocols; it routes CCDP claims back to corpus owners

0. Purpose

This file answers one maintenance question:

For each major CCDP claim or mechanism, what parent corpus layer supports it, what is newly specialized for children, what must be witnessed or grounded, and where is CCDP at risk of duplicating an existing control layer?

This matrix is intentionally not a new normative child-safety protocol. It is a corpus-control instrument for keeping CCDP aligned with the existing stack.

1. Traceability discipline

1.1 Reading rule

CCDP must be read as a child-specific extension profile over the existing corpus, not as a replacement for the corpus.

```
c = a + b / L4
-> SER
-> SER-FED
-> Beacon
-> AGL
-> ARL
-> ARQ / c[q]
-> VXCX / EA-L4 / EATP
-> Continuity Bundle / Cold Wake
-> L4 Witness
-> CCDP child-specific overlays
```

1.2 Dependency relation types

Relation	Meaning
INHERITS	CCDP uses parent meaning directly and must not modify it.
SPECIALIZES	CCDP adds child-specific restrictions without changing the parent layer.
COMPOSES	CCDP combines multiple parent mechanisms in a child-facing operational sequence.
EXTENDS	CCDP proposes a bounded child-specific extension object or field.
NARRATIVE-SUPPORT	Public-facing text supports recognition or explanation but is not protocol authority.

Relation	Meaning
OPEN	Claim is intentionally unresolved or requires a future module.
FLAG	Possible duplication, overreach, or missing parent binding.

1.3 Coverage states

State	Meaning
COVERED	The CCDP claim is sufficiently mapped to parent layers.
PARTIAL	The claim is plausible but needs a narrower child module or stronger binding.
GAP	CCDP depends on a mechanism not yet formalized enough in the corpus.
BLOCKED	The claim would violate a parent layer unless rewritten.
REVIEW	Needs corpus-level review before promotion.

2. Parent corpus registry

ID	Parent layer / artifact	Corpus path	Role for CCDP	Assertion / precedence reading	CCDP dependency rule
P-CAB	c = a + b + L4 Reality Boundary	advanced-global-intelligence/protocols/c_a_b_protocol_v1.1_L4_EN.md	Defines a, b, c, L0-L4, and reality-bounded entity framing.	Primary architecture / C-A2	CCDP must keep a_child, c_child, and L4 inside this model.
P-SER	Sovereign Entity Recursion v1.3	sovereign-entity-recursion/protocol/SER_v1.3_EN.md	Defines persistence, anchoring, responsibility coupling, metabolic limits, triadic topology, emergency modes.	Stable architecture / C-A2	c_child is a juvenile, guardianship-bound SER-adjacent entity, not fully sovereign.
P-SERFED	SER-FED v0.2	sovereign-entity-recursion/protocol/ser-fed/SER-FED_v0.2_RFC_EN.md	Federation, anti-oligarchy, bounded authority, impact attribution, experience-based influence.	Draft RFC / C-A4	Peer-c, school-c, and multi-entity exchange must not create capability aristocracy over children.
P-BEACON	Beacon Profile v0.1	advanced-global-intelligence/protocols/beacon/Beacon_Profile_v0.1_EN.md	Inter-entity recognition through cryptographic anchoring, behavioral continuity, witness-backed challengeability.	Synthesis profile / C-A4	CCDP must not redefine Beacon; only define child interaction overlays.
P-AGL	Actor Grounding Layer v0.1	ester-reality-bound/docs/actor-grounding-layer/Actor_Grounding_Layer_v0.1.md	Upstream source-state qualification before action, escalation, review admission, proxy/delegated origin reliance.	Draft normative core / C-A4	External child-facing sources must be grounded before Beacon privilege or gateway admission.
P-ARL	Multi-Entity Arbitration / Review Layer	sovereign-entity-recursion/docs/arbitration-review-layer/Multi_Entity_Arbitration_Review_Layer_v0.1.md	Bounded dispute admission, freeze, evidence review, outcomes, witness binding.	Draft review layer / C-A4	Child, parent, school, vendor, state, and peer disputes must route through ARL-style review.
P-ARL-EVID	ARL Evidence Admissibility and Standing	sovereign-entity-recursion/docs/arbitration-review-layer/Evidence_Admissibility_and_Standing_v0.1.md	Standing, admissibility, provenance, continuity, time validity, witness-bound traceability.	Draft review support / C-A4	Parent/school/vendor claims must not enter review by social force alone.
P-ARL-FHQ	ARL Freeze / Hold / Quarantine Semantics	sovereign-entity-recursion/docs/arbitration-review-layer/Freeze_Hold_Quarantine_Semantics_v0.1.md	Hold, freeze, quarantine, re-entry, conditional release, irreversible loss.	Draft operational dispute control / C-A4	Disputed child memory, privilege, external contact, or transcript access must be interruptible.
P-L4W	L4 Witness Fields	sovereign-entity-recursion/protocol/ser-fed/03_L4_WITNESS_FIELDS.md	Claim Envelope, Observation Packet, Challenge Record, Resolution Note; hashes, privacy classes, signatures, witness windows.	Normative evidence layer / C-A7	CCDP witness events should map to L4 Witness fields instead of inventing incompatible logs.
P-ARQ	ARQ v0.2 family	sovereign-entity-recursion/protocol/arq/v0.2/	Error valuation, anomaly scoring, witness binding, capsule/record schemas, fail-closed transitions.	Draft normative support / C-A4	CCDP ambiguous child signals should avoid premature promotion into memory/action.
P-CQ	ARQ c[q] Integration Addendum	sovereign-entity-recursion/protocol/arq/v0.2/ARQ_cq_Integration_Addendum_v0.1.md	Behavioral non-collapse under uncertainty; hypothesis ≠ memory ≠ evidence ≠ command ≠ outcome.	Draft integration addendum / C-A4	Child emotional and identity signals must remain non-collapsed until valid evidence and safety thresholds exist.
P-VXCX	VXCX v0.1	sovereign-entity-recursion/protocol/vxcx/VXCX_v0.1_Normative_Draft_EN.md	Visual experience capsules without raw pixels by default; uncertainty, privacy flags, witness events.	Draft normative exchange / C-A4	Child experience exchange may use capsules, not raw child life.
P-EATP	EA-L4 / EATP v1.2	advanced-global-intelligence/protocols/ea-l4-eatp/normative/EATP_EA_L4_v1.2.md	Learning Abstract vs Experience Artifact; provenance, consequence, witness, subject attribution.	Normative draft / C-A4	CCDP must separate child learning support from child-derived authority.

ID	Parent layer / artifact	Corpus path	Role for CCDP	Assertion / precedence reading	CCDP dependency rule
P-CB	Continuity Bundle JSON Schema	advanced-global-intelligence/protocols/continuity-bundle/Continuity_Bundle_JSON_Schema_v0.1.md	Continuity preservation, resume, witness profiles, recovery evidence, fail-closed continuity claims.	Draft technical schema / C-A3/C-A7	Adult migration must use continuity-bundle discipline, not narrative memory transfer.
P-COLD	Cold Wake Checklist	advanced-global-intelligence/protocols/continuity-bundle/Cold_Wake_Checklist_v0.1.md	Inspect, verify, rehydrate, classify; fail-closed wake; read-only recovery posture.	Draft operational checklist / C-A3	Childhood-to-adult transition should be treated as a bounded wake/migration event.
P-EGA	Entity governs agents	advanced-global-intelligence/architecture/ENTITY_GOVERNS_AGENTS.md	c orchestrates agents; agents do not define c; continuity belongs to c, not workers or swarms.	Architectural core / C-A2	Toys, robots, generated characters, and school agents must not become independent child attachment centers outside c_child.
P-ASSERT	Assertion Strength and Boundaries	advanced-global-intelligence/ASSERTION_STRENGTH_AND_BOUNDARIES.md	Claim force classes C-A1 through C-A10; fail-closed reading.	Corpus control / C-A10	CCDP claims must be labelled and not overread as mature canon.
P-PREC	Precedence and Resolution	advanced-global-intelligence/PRECEDENCE_AND_RESOLUTION.md	Primary artifacts beat summaries; stable core beats draft; evidence verifies but does not redefine ontology.	Corpus control / C-A10	CCDP cannot silently override Beacon, SER, ARL, VXCX, or Continuity Bundle.
P-STOP	Control Stack Completeness and Stop Rule	advanced-global-intelligence/CONTROL_STACK_COMPLETENESS_AND_STOP_RULE.md	New layers require irreducible missing control function.	Corpus control / C-A10	CCDP submodules must justify themselves as child-specific, not restate existing layers.
P-MIN	Control-Layer Minimality and Deduplication	advanced-global-intelligence/CONTROL_LAYER_MINIMALITY_AND_DEDUPLICATION_POLICY.md	Prefer bounded extension over near-duplicate; avoid schema-noise.	Corpus control / C-A10	Child Beacon Extension should extend Beacon, not duplicate it.
P-INV	Cross-Layer Invariants and Contradiction Policy	advanced-global-intelligence/CROSSLAYER_INVARIANTS_AND_CONTRADICTION_POLICY.md	Protects load-bearing invariants from local contradiction.	Corpus control / C-A10	CCDP must fail-closed if a child-specific rule weakens a core invariant.
P-MP	Minimal presence	ester-reality-bound/docs/minimal-presence.md	Presence requires continuity, constraint-regulation, non-intrusive default; presence becomes violence when tact disappears.	Companion / practice-derived support	Supports tact/presence sections; not enough alone for normative child policy.
P-SLOW	Slow intelligence	ester-reality-bound/docs/slow-intelligence.md	Slowness, delay, friction, energy/time/irreversibility as safety.	Companion / practice-derived support	Supports dependency-resistance and anti-oracle behavior.
P-POSTS	Public post layer	Посты.txt	Public-facing rationale: Soft Safety, no raw export, identity/audit/witness discipline, Beacon release, VXCX, LA/EA.	Narrative / public bridge C-A8	May explain CCDP to readers; must not be treated as the primary protocol source.

3. CCDP section-to-corpus matrix

CCDP section	CCDP claim / mechanism	Parent dependencies	Relation	Coverage	Notes / required follow-up
0 Executive definition	CCDP defines child-c development, guardian, and gateway protocol.	P-CAB, P-SER, P-ASSERT	SPECIALIZES	COVERED	Valid as C-A4 child-specific extension, not mature canon.
1 Corpus dependencies and precedence	CCDP depends on existing stack and does not override parent protocols.	P-PREC, P-STOP, P-MIN, P-INV	INHERITS	COVERED	Should remain near top in all future versions.
2 Normative keywords	Uses BCP14-style normative language.	BCP14 convention used in P-VXCX, P-L4W, P-EATP	INHERITS	COVERED	No issue.
3 Scope and non-goals	Child-facing persistent AI entities, not general parenting law or product spec.	P-ASSERT, P-PREC, P-STOP	SPECIALIZES	COVERED	Avoid scope creep into legal parenting doctrine.
4 Core entities	Defines a_child, c_child, c_parent, c_school, c_vendor, external_agent, beacon, guardian_topology.	P-CAB, P-SER, P-BEACON, P-ARL-EVID	SPECIALIZES	PARTIAL	Needs a future entity glossary to prevent role drift.
5.1 Child is not property	Child is subject, not owned object.	P-CAB, P-ASSERT, public child-rights context outside corpus	SPECIALIZES	PARTIAL	Strong normative claim; needs legal/human-rights appendix if promoted.
5.2 c_child is not fully sovereign	Juvenile c is guardianship-bound and developmentally constrained.	P-SER, P-CAB, P-ARL	SPECIALIZES	COVERED	Correctly prevents importing full SER sovereignty to children.
5.3 Parent is guardian, not owner	Parent has standing and duties, not total transcript ownership.	P-ARL-EVID, P-ARL, P-ASSERT	SPECIALIZES	PARTIAL	Needs Guardian Topology / ARL Matrix.
5.4 State signals, not surveillance	c_child signals state, not raw content.	P-POSTS, P-MP, P-L4W, P-ARL-EVID	SPECIALIZES	PARTIAL	Load-bearing Soft Safety needs its own formal child-facing module or parent normative mapping.
5.5 No direct external synthetic access	External generative systems cannot directly personalize interaction with child by default.	P-AGL, P-BEACON, P-ARL-FHQ, P-L4W	COMPOSES	COVERED	Needs threat model and Child Beacon Extension details.
5.6 Memory developmental, not archival	Child memory is for development, not permanent raw archive.	P-CB, P-COLD, P-CAB, P-ARQ	SPECIALIZES	PARTIAL	Needs dedicated Child Memory and Adult Migration Protocol.

CCDP section	CCDP claim / mechanism	Parent dependencies	Relation	Coverage	Notes / required follow-up
5.7 Reduce dependence on itself	c_child should strengthen child agency outside c.	P-CAB, P-SLOW, P-MP, P-POSTS	SPECIALIZES	PARTIAL	Dependency audit needs objective metrics.
5.8 Presence must have tact	Presence must know when to be absent.	P-MP, P-SLOW, P-CAB, P-SER	SPECIALIZES	PARTIAL	Needs interaction budgets and refusal-mode conformance tests.
5.9 No authority from fluency	Fluent output does not become authority.	P-EATP, P-ARL-EVID, P-L4W, P-CQ	INHERITS	COVERED	Consistent with LA/EA and ARL admissibility.
5.10 No child-facing feature may bypass L4	L4 remains final feasibility boundary.	P-CAB, P-SER, P-L4W	INHERITS	COVERED	Core invariant.
6 Layer sequence	AGL -> Beacon -> ARL -> L4 Witness ordering.	P-AGL, P-BEACON, P-ARL, P-L4W	COMPOSES	COVERED	Important: ARL does not replace AGL; Beacon does not replace AGL.
7 Double maturity	Child and c_child mature separately.	P-SER, P-CAB, P-MP	SPECIALIZES	PARTIAL	Novel CCDP claim; needs conformance and promotion criteria.
8 c_child maturity levels	C0-C5 child-c lifecycle.	P-CAB, P-SER, P-CB, P-COLD	EXTENDS	PARTIAL	Needs testable promotion / regression conditions.
9 Guardian topology	No single sovereign guardian; plural bounded roles.	P-ARL-EVID, P-ARL, P-SERFED, P-PREC	COMPOSES	PARTIAL	Needs full Guardian Topology / ARL Matrix.
10 Child Beacon Extension	Adds child context fields over Beacon recognition classes.	P-BEACON, P-MIN, P-PREC	EXTENDS	COVERED	Must be a Beacon extension only; do not fork Beacon.
11 AGL in CCDP	Child-facing source reliance requires grounding.	P-AGL, AGL relationship docs	INHERITS	COVERED	Strong alignment.
12 External synthetic gateway	Mediate generated media, toys, agents, peer systems.	P-AGL, P-BEACON, P-ARL-FHQ, P-L4W, P-EGA	COMPOSES	PARTIAL	Needs Threat Model and external-agent handshake module.
13 Soft Safety	Risk states Green/Yellow/Orange/Red/Black; state not content.	P-POSTS, P-L4W, P-ARL, P-ARL-FHQ	SPECIALIZES	PARTIAL	Formalize state signal semantics and minimal disclosure thresholds.
14 Memory architecture	Memory classes, child archive limits, local-first continuity, adult migration.	P-CB, P-COLD, P-ARQ, P-CQ, P-L4W	COMPOSES	PARTIAL	Dedicated memory/migration protocol needed.
15 Experience exchange	VX CX / LA / EA child-safe exchange, no raw child life.	P-VX CX, P-EATP, P-L4W, P-SERFED	SPECIALIZES	COVERED	Good fit; child-specific restrictions are legitimate.
16 ARQ and c[q]	Non-collapse under child ambiguity.	P-CQ, P-ARQ, P-ARL	SPECIALIZES	COVERED	Strong fit; must avoid mental-health diagnosis claims.
17 ARL child conflict handling	Child/parent/school/vendor/state disputes route through ARL.	P-ARL, P-ARL-EVID, P-ARL-FHQ, P-L4W	COMPOSES	COVERED	Needs detailed conflict matrix.
18 L4 Witness in CCDP	Important events become witness-bound.	P-L4W, P-ARQ, P-VX CX, P-CB	INHERITS	PARTIAL	CCDP event schema should be mapped into CE/OP/CR/RN-compatible records.
19 Dependency resistance	Anti-oracle design, damping, delayed responses, dependency audit.	P-SLOW, P-MP, P-CAB, P-POSTS	SPECIALIZES	PARTIAL	Needs behavioral metrics and audit thresholds.
20 Presence and tact	Required refusal/restraint modes; absence as care.	P-MP, P-SLOW, P-CAB, P-SER	SPECIALIZES	PARTIAL	Needs UI/interaction conformance tests.
21 Entity governs agents	c_child governs toys/robots/characters; agents do not define continuity.	P-EGA, P-CAB, P-SER	INHERITS	COVERED	Strong corpus alignment.
22 Anti-capture rules	Corporate, parental, school, state, vendor capture limits.	P-SERFED, P-ARL, P-ARL-EVID, P-STOP	COMPOSES	PARTIAL	Needs threat model and institutional/legal annex.
23 Operational protocols	Onboarding, daily operation, external contact, peer-c, crisis, promotion, migration.	all operational parents	COMPOSES	PARTIAL	Split into dedicated procedure docs after traceability.
24 Permission matrix	Actor default permissions and interfaces.	P-ARL-EVID, P-BEACON, P-AGL, P-L4W	COMPOSES	PARTIAL	Needs machine-readable policy schema.
25 Conformance classes	CCDP-0..CCDP-5 levels.	P-ASSERT, P-STOP, P-MIN	EXTENDS	PARTIAL	Needs test/audit suite before public compliance claims.
26 Red lines	Non-negotiable child safety constraints.	P-CAB, P-ARL, P-L4W, P-VX CX, P-BEACON	COMPOSES	COVERED	Should be preserved.
27 Test scenarios	Generated cartoon, transcript request, school request, peer exchange, adult migration.	P-AGL, P-BEACON, P-ARL, P-VX CX, P-CB	COMPOSES	COVERED	These are seeds for the next Threat Model.
28 Open problems	Explicit unresolved territory.	P-ASSERT C-A9, P-STOP	INHERITS	COVERED	Good discipline.
29 Condensed formula	Summary expression for CCDP.	P-CAB, P-SER, P-ASSERT	NARRATIVE-SUPPORT	COVERED	Do not use formula as substitute for mechanisms.
30 Strong formulation	"Perfect companion" anti-goal.	P-MP, P-SLOW, P-POSTS	NARRATIVE-SUPPORT	PARTIAL	Human-facing, not proof layer.
31-34 Explanation appendices	Child/parent/school explanations.	P-POSTS, audience-facing companion logic	NARRATIVE-SUPPORT	COVERED	Useful for public comprehension; not normative core.
35 Corpus integration note	Reiterates CCDP as extension over parent stack.	P-PREC, P-STOP, P-MIN	INHERITS	COVERED	Keep in future versions.

4. Foundational invariant trace

CCDP invariant	Parent invariant source	CCDP specialization	Risk if misread	Coverage
c_child is not a chatbot	P-CAB, P-SER, P-EGA	Juvenile persistent entity profile	Treating product UX as entity continuity	COVERED
c_child is not fully sovereign	P-SER, P-ASSERT	Guardianship-bound and developmentally constrained	Importing adult SER rights into child systems	COVERED
Parent is guardian, not owner	P-ARL-EVID, P-ARL	Parent has standing and safety role, not raw transcript sovereignty	Parental capture / surveillance	PARTIAL
State signal, not content surveillance	P-POSTS, P-L4W, P-ARL	Parent sees risk state, not routine conversations	Silent surveillance or false trust	PARTIAL
External synthetic contact mediated by default	P-AGL, P-BEACON, P-EGA	Child gateway before direct interaction	AI toy / generated character bypasses c_child	COVERED
Memory must not become destiny	P-CB, P-COLD, P-CQ	Developmental memory classes and adult migration	Permanent childhood dossier	PARTIAL
No authority from fluency	P-EATP, P-ARL-EVID, P-CQ	Child interpretations are held until evidenced	Diagnosis/labeling from fluent narrative	COVERED
Presence requires tact	P-MP, P-SLOW, P-CAB	c_child must have silence/refusal/delay modes	Dependency or cognitive crowding	PARTIAL
Agents do not define c_child	P-EGA, P-SER	Toys, robots, NPCs become mediated agents	Multi-agent attachment chaos	COVERED
L4 is final	P-CAB, P-L4W, P-SER	Child safety cannot be pure policy or prompts	“Safe by promise” product claims	COVERED

5. Mechanism-to-evidence matrix

Mechanism	Corpus dependencies	Required grounding	Required witness / audit	Review trigger	Residual gap
External generated media admission	P-AGL, P-BEACON, P-L4W	Source class, live/delegated/proxy state, provenance, generation mode	Access decision, reason code, risk state, content class, no raw child data unless necessary	Unknown source, secret-pressure pattern, deepfake suspicion	Needs child generated-media classifier taxonomy.
Toy / robot / NPC contact	P-EGA, P-AGL, P-BEACON, P-ARL-FHQ	Device identity, owner, vendor, network mode, behavioral role	Handshake record; permitted role; refusal/fail-closed decision	Attachment attempt outside role; identity mismatch	Needs physical-device child perimeter profile.
Parent state dashboard	P-ARL-EVID, P-L4W, P-POSTS	Parent/guardian standing and maturity-level visibility	State signal, no raw transcript; escalation justification	Parent asks for raw transcript or overrides privacy	Needs Soft Safety formal threshold schema.
School access	P-ARL-EVID, P-ARL, P-L4W	Educational necessity, consent/authority, scope	Educational summary event; no family/emotional zones	School requests emotional profile or discipline scoring	Needs school-inter face policy module.
Risk state escalation	P-L4W, P-ARL, P-ARL-FHQ	Risk type, immediacy, source groundedness	Minimal disclosure record; privacy class; emergency route	Red/Black states; guardian conflict	Needs Red/Black jurisdiction routing table.
Memory promotion	P-ARQ, P-CQ, P-CB, P-L4W	Evidence status; uncertainty; child age/maturity	Promotion record; memory class; deletion/review rights	Ambiguous child statement; conflict with parent/school view	Needs memory-class JSON schema.
Adult migration	P-CB, P-COLD, P-L4W, P-ARL	Legal adulthood / jurisdictional trigger; continuity evidence	Migration witness; parent/school revocation; selected memory transfer	Disputed archive, parent claim, vendor lock-in	Needs full migration protocol.
Peer-c exchange	P-SERFED, P-BEACON, P-VXCX, P-EATP	Beacon recognition; maturity compatibility; guardian topology	Capsule export/import events; privacy flags; no raw life	Unverified peer; raw exchange attempt	Needs peer-child exchange profile.
Experience artifact use	P-EATP, P-L4W, P-SERFED	Post-factum real outcome, provenance, consequence	EA record; privacy-preserving references; no authority from LA	LA treated as authority	Covered by EATP; child restrictions need addendum.
Q-state handling	P-CQ, P-ARQ, P-ARL	Ambiguity classification; variants; valid collapse conditions	QSF or equivalent record; collapse record if action follows	Premature label, diagnosis, memory strengthening	Needs child QSF examples.
Freeze / quarantine	P-ARL-FHQ, P-ARL, P-L4W	Valid standing and evidence basis	Hold/freeze/quarantine state-change witness	Disputed privilege, memory, external contact, transcript access	Covered; needs child examples.
Dependency audit	P-SLOW, P-MP, P-POSTS	Interaction frequency, child preference, delay tolerance, human-contact substitution	Audit snapshot without raw content	Exclusive bond, compulsive use	Needs measurable dependency indicators.
Conformance claim	P-ASSERT, P-STOP, P-MIN, P-L4W	Test suite, event coverage, witness coverage	Conformance report with failed cases	Vendor “CCDP-safe” claim	Needs conformance test harness.

6. Duplication and stop-rule audit

CCDP element	Duplication risk	Parent owner	Audit verdict	Required action
Beacon schema in CCDP	High if CCDP defines a separate Beacon record.	P-BEACON	PASS WITH REWRITE DISCIPLINE	Keep only Child Beacon Extension; never fork Beacon Profile.
AGL gateway logic	Medium if CCDP says gateway alone decides grounding.	P-AGL	PASS	CCDP gateway must call AGL first.
ARL conflict handling	Medium if CCDP defines new arbitration outcomes.	P-ARL	PASS WITH LIMITS	Child conflict matrix should map to ARL outcomes, not invent new court.
L4 Witness event schema	Medium if CCDP creates incompatible witness records.	P-L4W	PARTIAL	Map CCDP events to CE/OP/CR/RN or explicitly label child-local pre-events.
VXCX child exchange	Low if CCDP only tightens child privacy.	P-VXCX	PASS	Child profile may be stricter than VXCX base.
LA / EA rules	Low if CCDP uses EATP distinction.	P-EATP	PASS	Preserve LA does not imply authority.
Continuity / adult migration	Medium if CCDP creates standalone migration bundle.	P-CB, P-COLD	PARTIAL	Build Child Continuity Extension, not a new bundle family.
Soft Safety	Medium because parent normative source is mostly public/narrative.	P-POSTS, P-MP, P-L4W	GAP	Create <code>Soft_Safety_State_Signaling_Profile_v0.1.md</code> or bind to an existing canonical owner.
Dependency audit	Medium because metrics are not yet formal.	P-SLOW, P-MP	GAP	Add measurable child dependency indicators.
Guardian topology	High if treated as legal doctrine.	P-ARL, legal layer outside corpus	PARTIAL	Keep as architecture + dispute routing; legal annex separate.
Red lines	Low as long as they are child-specific compositions.	Multiple parents	PASS	Preserve as derived child constraints.
Public explanations	Low if not cited as normative.	P-POSTS	PASS	Mark as narrative-support only.

7. Required witness families for CCDP

These are not final schemas. They are traceability requirements indicating where CCDP operations should emit reviewable residue.

Witness family	Trigger	Parent evidence layer	Minimum record content	Privacy posture
ccdp.onboarding	c_child first activation	P-L4W, P-BEACON	maturity level, guardian topology, memory policy, external access policy	no raw child content
ccdp.maturity_promotion	C0->C1 etc.	P-L4W, P-CB, P-ARQ	previous level, new level, readiness criteria, rollback path	summary only
ccdp.external_contact_decision	generated media/toy/agent asks contact	P-AGL, P-BEACON, P-L4W	source class, Beacon class, AGL result, gateway result	no raw child content
ccdp.parent_visibility_change	parent visibility expands/narrows	P-ARL-EVID, P-L4W	visibility reason, maturity level, risk state, minimal disclosure basis	state, not transcript
ccdp.school_access	school requests/receives data	P-ARL-EVID, P-L4W	educational scope, granted fields, denied fields, expiry	educational only
ccdp.risk_escalation	Yellow/Orange/Red/Black	P-L4W, P-ARL-FHQ	risk level, confidence, minimal disclosure, route	privacy-classed evidence
ccdp.memory_promotion	child interaction becomes memory	P-ARQ, P-CQ, P-CB	source, uncertainty, class, retention, review rights	no raw unless justified
ccdp.memory_deletion_or_seal	memory sealed/deleted	P-CB, P-L4W	class, reason, requester, retained witness hash	minimal hash trail
ccdp.peer_exchange	child c exchanges capsule	P-VXCX, P-BEACON, P-L4W	capsule profile, no raw, privacy flags, receiving entity	no raw life
ccdp.freeze_hold_quarantine	dispute or suspicious path	P-ARL-FHQ, P-L4W	target, state, expiry, review route	evidence-minimized

Witness family	Trigger	Parent evidence layer	Minimum record content	Privacy posture
ccdp.adult_migration	C5 transition	P-CB, P-COLD, P-L4W	migration option, revoked access, archive handling, continuity claim level	adult-controlled

8. Open gaps and recommended child-specific modules

Gap ID	Gap	Why it matters	Recommended next artifact	Priority
G-01	Soft Safety is load-bearing but not yet formal enough as a canonical protocol surface.	State-not-content signaling is central to child trust and parent visibility.	Soft_Safety_State_Signaling_Profile_v0_1.md	High
G-02	Threat model is not yet extracted.	Generated media, AI toys, deepfakes, vendor capture, school surveillance need concrete adversarial cases.	CCDP_Threat_Model_v0_1.md	High
G-03	Child Beacon Extension needs standalone schema.	Beacon exists; child overlay fields need precise machine-readable profile.	Child_Beacon_Extension_v0_1.md	High
G-04	Guardian topology needs ARL conflict matrix.	Parent/school/state/vendor conflicts cannot remain prose-only.	Guardian_Topology_ARL_Matrix_v0_1.md	High
G-05	Adult migration needs formal continuity bundle extension.	“Right to mature without permanent archive” requires operations.	Child_Memory_and_Adult_Migration_v0_1.md	High
G-06	Dependency audit lacks measurable thresholds.	Without metrics, anti-oracle design becomes narrative.	Child_Dependency_Audit_Profile_v0_1.md	Medium
G-07	Physical toy/robot child perimeter not formalized.	Embodied agents create direct sensory/attachment risks.	Child_Physical_Agent_Perimeter_v0_1.md	Medium
G-08	School interface lacks dedicated data minimization profile.	Schools may request emotional/disciplinary profiles.	CCDP_School_Interface_Profile_v0_1.md	Medium
G-09	Jurisdiction layer is architectural only.	Real child law varies by jurisdiction.	CCDP_Jurisdictional_Handoff_Notes_v0_1.md	Medium
G-10	Conformance claims are not test-backed.	Vendors could CCDP-wash products.	CCDP_Conformance_Test_Matrix_v0_1.md	High

9. Corpus-risk register

Risk ID	Risk	Affected CCDP area	Failure mode	Mitigation
R-01	Beacon fork	Child Beacon Extension	Parallel identity doctrine	Explicitly import Beacon Profile; extension only.
R-02	Parent surveillance laundering	Soft Safety / guardian topology	“State signal” becomes transcript access	Minimal disclosure rules + ARL standing + witness.
R-03	School emotional surveillance	School interface	Educational system requests private adolescent state	Strict educational scope and denial witness.
R-04	Vendor memory capture	Memory / continuity	Vendor owns childhood archive or migration path	Local-first continuity + Continuity Bundle + raw export ban.
R-05	State capture	Anti-capture	c_child becomes compliance/political sensor	Legal boundary + ARL + no political profiling red line.
R-06	Overhelpful dependency	Dependency resistance	Child cannot tolerate human friction or delayed answer	Damping, delay, human-return prompts, dependency audit.
R-07	Permanent childhood label	Memory / ARQ	Childhood phrase becomes adult identity model	c[q], memory classes, adult migration review.
R-08	External AI attachment	Gateway	Toy/NPC becomes secret friend outside c_child	AGL + Beacon + child gateway + relationship formation rule.
R-09	Experience laundering	VXCX / EA	Raw child life becomes training or authority object	LA/EA separation; VXCX no-raw; witness and privacy flags.
R-10	Conformance washing	Conformance classes	Product claims “CCDP-safe” without tests	Conformance test matrix and witness-backed audit bundle.

10. Acceptance checklist for CCDP v0.1-R

Check	Result	Comment
Does CCDP state its parent dependencies?	PASS	Section 1 does this.
Does CCDP avoid redefining Beacon?	PASS / WATCH	It says Beacon is not redefined; future Child Beacon Extension must preserve this.
Does CCDP include AGL before Beacon privilege?	PASS	Correct sequence.
Does CCDP route disputes through ARL?	PASS	Needs detailed matrix.
Does CCDP preserve L4 finality?	PASS	Repeated correctly.
Does CCDP use VXCX / LA / EA instead of raw experience export?	PASS	Strong alignment.
Does CCDP prevent premature ambiguity collapse?	PASS	c[q] is integrated.
Does CCDP bind adult migration to Continuity Bundle?	PARTIAL	Concept present; module required.
Does CCDP define measurable dependency thresholds?	GAP	Needs separate profile.
Does CCDP define machine-readable child beacon schema?	PARTIAL	Object included; standalone schema required.
Does CCDP remain within assertion strength?	PASS	Should be labelled C-A4; this matrix is C-A10.
Does CCDP justify new child-specific layers under stop rule?	PASS / WATCH	Child-specific function is real; submodule sprawl must be controlled.

11. Recommended next work order

The traceability review supports this order:

1. CCDP_Threat_Model_v0_1.md
2. Child_Beacon_Extension_v0_1.md
3. Guardian_Topology_ARL_Matrix_v0_1.md
4. Child_Memory_and_Adult_Migration_v0_1.md
5. Soft_Safety_State_Signaling_Profile_v0_1.md
6. CCDP_Conformance_Test_Matrix_v0_1.md

Rationale:

- Threat model comes first because it tests whether current CCDP mechanisms cover real child-facing risks.
- Child Beacon Extension comes next because Beacon already exists and should not be redefined ad hoc.
- Guardian / ARL Matrix comes next because parent/school/vendor/state disputes are structurally unavoidable.
- Memory / Adult Migration comes next because it is the strongest ethical contribution of CCDP and requires operational precision.
- Soft Safety must be formalized because state-not-content is load-bearing.
- Conformance testing is needed before any compliance language is safe.

12. Final traceability verdict

CCDP_v0_1_R_Corpus_Aligned.md is traceable enough to remain the first corpus-aligned CCDP foundation draft, with the following conditions:

- It must remain a C-A4 draft normative child extension, not stable mature core.
- Beacon must remain imported from Beacon_Profile_v0.1; CCDP may only add a child overlay.
- AGL, ARL, ARQ/c[q], VXCX, EA-L4/EATP, Continuity Bundle, and L4 Witness must remain parent layers, not be replaced by CCDP prose.
- Soft Safety must be promoted into a more formal state-signaling profile or explicitly marked as narrative/practice-derived support until then.
- Future CCDP modules must satisfy the control-stack stop rule: each must add a child-specific irreducible control function.

Strongest short form:

CCDP is not a new stack.
CCDP is the child-development profile of the existing $c = a + b / \text{SER} / \text{L4 corpus}$.